SIGMA-Net

Satellite-grounded Inference for Geospatial Map Anchoring

Technical Report — ESA Φ-lab x ITU x AI for Good: *Reaching New Heights with GeoFM*

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Final Score	IoU Building	IoU Vegetation	IoU Water	RMSE Bld. Ht.	RMSE Veg. Ht.
	0.6277	0.9193	0.7129	1.608 m	2.868 m

Abstract

*We present SIGMA-Net, a two-stage residual correction framework for building/vegetation cover segmentation and height estimation from AlphaEarth satellite embeddings. Rather than solving the full multi-task prediction problem from scratch (the master problem), we first train a dedicated Geospatial Foundation Model — GeoMTConvNeXt — to produce a strong structural prior. A compact 9.2M-parameter corrector, SIGMA-Net, then learns to detect and fix the errors in that prior (the slave problem), achieving a public leaderboard score of **0.6030**.*

1. Problem Framing: Master vs. Slave Problem

The master problem — predicting five per-pixel quantities from compressed 64-channel satellite embeddings — is hard for three compounding reasons. **First**, the ground-truth labels are derived from airborne LiDAR surveys. For buildings specifically, this introduces rooftop measurement noise from complex roof structures, temporal misalignment between LiDAR acquisition and satellite data, point-cloud density variation on reflective glass facades (producing systematically underestimated height labels), and DTM subtraction artefacts on sloped terrain. **Second**, the 64-channel compressed embeddings discard the fine-grained boundary detail most critical for building footprint delineation, forcing a model to decode the latent structure of the embedding space before it can learn the prediction task. **Third**, the five tasks have fundamentally incompatible loss surfaces: building cover is a sparse binary problem (1-5% foreground), vegetation is near-saturated, and height is a heavy-tailed regression — competing gradients prevent joint convergence.

We replace the master problem with the structurally simpler **slave problem**: given a mostly-correct prior map, identify where it is wrong and correct it.

	Master Problem	Slave Problem (SIGMA-Net)
Output space	Unbounded per-pixel	Bounded residual: Delta-Cover in [-0.05,+0.15], Delta-h in [-2,+5] m
Must learn	Geography + LiDAR biases + spectral decoding	Error patterns of a known prior
Capacity needed	Large pretrained GFM	9.2M-parameter corrector
Signal quality	Dense but LiDAR-noisy	Binary error masks + signed residuals

2. Stage 1 — GeoMTConvNeXt: Pretraining a Geospatial Foundation Prior

Before training SIGMA-Net, we trained a dedicated multi-task segmentation and height estimation model — **GeoMTConvNeXt** — from scratch on the full training set. The purpose was explicit: *to produce the richest possible structural prior from the AlphaEarth embeddings, so that Stage 2 only needs to operate on a compact residual rather than solving the full prediction problem cold*.

Architecture

GeoMTConvNeXt is a **ConvNeXt UNet**: a 4-level encoder-decoder with skip connections, where every block follows the ConvNeXt design — depthwise 7x7 convolution, BatchNorm, GELU activation, and 4x pointwise channel expansion. The encoder progressively downsamples the 64-channel AlphaEarth input from 256x256 down to a 16x16 bottleneck, building rich multi-scale feature representations at each level. The decoder reconstructs spatial resolution through bilinear upsampling and concatenated skip connections from the corresponding encoder level, recovering fine-grained spatial detail lost during downsampling. Four independent prediction heads branch from the decoder

output: three sigmoid heads for building, vegetation, and water cover (one channel each), and one linear head for height estimation.

Loss and Training Objective

The model is trained jointly on all five tasks with a composite loss. Cover segmentation uses a **Lovasz IoU surrogate** combined with binary cross-entropy — the Lovasz term directly optimises the IoU metric and is critical for the building and water classes where foreground pixels are rare. Height is supervised with a **masked SmoothL1 loss** with 2x weight on building pixels, reflecting their higher importance in the competition scoring formula (0.25 weight vs 0.20 for vegetation height).

Why Pretrain Separately

Training GeoMTConvNeXt as an independent stage rather than jointly with SIGMA-Net serves two purposes. First, it decouples the hard problem of learning the AlphaEarth embedding space from the correction task, allowing each model to specialise. Second, the prior's predictions become fixed, stable targets that SIGMA-Net can reliably learn to correct — a moving target from a jointly-trained backbone would destabilise the residual supervision signal for the six correction heads.

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3. Stage 2 — SIGMA-Net: Confidence-gated Residual Corrector

SIGMA-Net takes the AlphaEarth embedding (64 ch) and the Stage 1 prior (4 ch) as joint input (68 ch total) and learns to answer three questions per pixel, per task:

1. ***Is the prior wrong here?*** → *Discrepancy Localizer*
2. ***If wrong, by how much and in which direction?*** → *Residual Estimator*
3. ***How confident are we?*** → *Correction Confidence*

The same ConvNeXt UNet backbone processes the joint input. Six isolated prediction heads branch from the final decoder feature map:

Head	Output	Act.	Role
discrepancy_localizer	[B,3,H,W]	Sigmoid	Detect cover error pixels
residual_estimator	[B,3,H,W]	Tanh	Signed correction direction in [-1,+1]
correction_confidence	[B,3,H,W]	Sigmoid	Reliability of cover correction
height_discrepancy_localizer	[B,1,H,W]	Sigmoid	Detect height error pixels
height_residual_estimator	[B,1,H,W]	Tanh	Signed height correction direction
height_correction_confidence	[B,1,H,W]	Sigmoid	Reliability of height correction

Evidence Gate and Fusion

```
gate_c    = discrepancy_prob  x  correction_confidence
delta_cov = clamp(gate_c x residual x 0.15,   -0.05, +0.15)
cover_out = clamp(prior_cover + delta_cov,       0.0,   1.0)

gate_h      = h_discrepancy_prob  x  h_correction_confidence
delta_h     = clamp(gate_h x h_residual x 5.0,  -2.0, +5.0)
height_out  = clamp(prior_height + delta_h,       0.0,  60.0)
```

The triple gate (discrepancy x confidence x residual) ensures SIGMA-Net is *conservative by default*: where the heads do not agree, the prior passes through unchanged. The asymmetric height clamp [-2, +5] m reflects the LiDAR under-prediction bias on tall buildings described in Section 1.

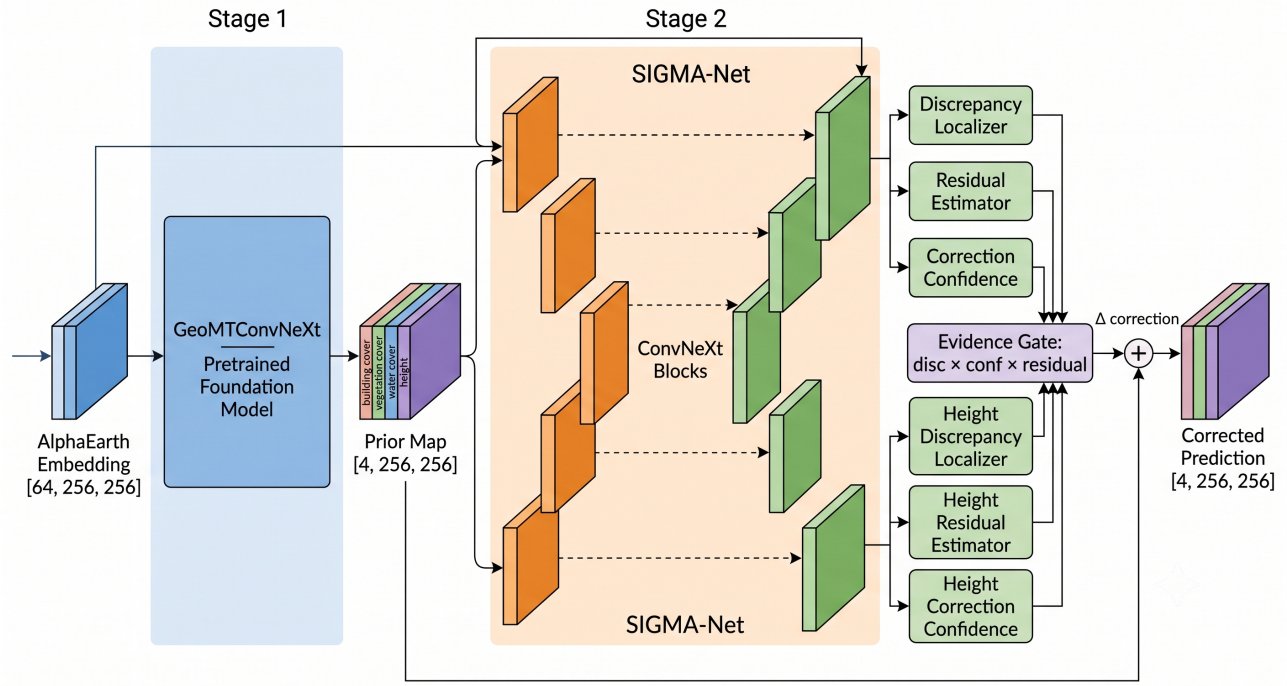


Figure 1 *— SIGMA-Net two-stage pipeline. Stage 1 (GeoMTConvNeXt, left) generates the 4-channel prior from the 64-ch AlphaEarth embedding. Stage 2 (SIGMA-Net, centre) fuses both through a ConvNeXt UNet and branches into six heads. The evidence gate (right) applies a bounded, confidence-gated correction to the prior.*

4. Training

Supervision targets for all six SIGMA-Net heads are derived automatically from the residual (GT - prior): discrepancy masks at |error| > 0.10 (cover) or > 1.0 m (height), normalized signed residuals for the estimators, and softer confidence masks at half those thresholds. No additional annotation is required beyond the competition labels.

Term	Formula	Purpose
L_cover	L1 + Lovasz IoU	Directly optimises the IoU metric
L_discrepancy	BCE	Trains localizer on error > 0.10 mask
L_residual	MSE (error regions only)	Trains estimator where corrections needed
L_confidence	BCE	Calibrates confidence head
L_tv	Total variation on residuals	Spatial smoothness of correction field
L_height	SmoothL1 (bld x2, veg x1.5) + head BCE/MSE	Height with building emphasis

Hyperparameter	Value
Optimizer	AdamW (lr=3e-4, wd=1e-4)
Scheduler	Cosine annealing, 40 epochs, eta_min=1e-6
Batch / Precision	8 / bfloat16 autocast
Validation	20% random hold-out; checkpoint gate: IoU_bld > 0.35

5. Results (Official Public Leaderboard)

Metric	Score
Final Score	0.6030

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IoU Building	0.6277
IoU Vegetation	0.9193
IoU Water	0.7129
RMSE Building Height	1.608 m
RMSE Vegetation Height	2.868 m

Vegetation IoU (0.9193) is near-saturated, reflecting GeoMTConvNeXt's strong prior on large continuous cover classes. Building IoU (0.6277) and building height RMSE (1.608 m) are the primary improvement levers — exactly where LiDAR label noise is highest and where SIGMA-Net's evidence gate applies the most targeted corrections.

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6. Conclusion

The master/slave decomposition — pretraining GeoMTConvNeXt to absorb the hard embedding-space decoding and LiDAR-label complexity, then training SIGMA-Net to correct only the residual errors — proves both principled and efficient. A 9.2M-parameter corrector trained in 40 epochs achieves **0.6030** on the public leaderboard, with conservative gating ensuring the prior is never degraded in regions it predicted correctly.